Supporting Information for: Metric-consistent neural networks for genomic prediction and selection

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This document (**File S1**) collects the full numerical results underlying the manuscript. Every value is generated directly from the deposited result files by `analysis/make_supplement.py`; no number is transcribed by hand. Throughout, "Δ" denotes the change of a metric relative to the mean-squared-error (MSE) baseline for an otherwise identical neural network (architecture and optimizer frozen across losses), summarized over (dataset, trait) cells with bias-corrected and accelerated (BCa) bootstrap 95% confidence intervals and Holm-corrected paired Wilcoxon signed-rank p-values. Unless stated otherwise, neural-network predictions are affine-calibrated. The proprietary sorghum case study is summarized in the main text and is not reproduced here.

Contents

1 Numerical verification of the equivalences (Experiment A)

Table S1 confirms, to machine precision, the algebraic identities that motivate the standardized-MSE Pearson loss: the standardized MSE equals $2(1-r)$ (Proposition 1), the affine-optimal MSE identity holds (Proposition 2), and the analytic loss gradients match automatic differentiation.

Table S1: Experiment A: numerical verification of the equivalences. Proposition 1 (standardized MSE equals $2(1-r)$) and Proposition 2 (the affine-optimal MSE identity) are confirmed to machine precision across randomized cases; analytic gradients of the Pearson and CCC losses pass `torch.autograd.gradcheck`; and the identity holds exactly on a real dataset. From `results/exp_a.json`.

Check	Quantity	Value	n
Prop. 1: standardized-MSE $= 2(1-r)$	max abs. diff.	5.02e-08	80
Prop. 1: standardized-MSE $= 2(1-r)$	mean abs. diff.	7.38e-09	80
Prop. 2: affine-optimal MSE identity	max abs. diff.	1.19e-15	200
Prop. 2: affine-optimal MSE identity	mean abs. diff.	1.80e-16	200
Gradient check (Pearson loss)	`gradcheck` passes	yes	–
Gradient check (CCC loss)	`gradcheck` passes	yes	–
Neg-Pearson gradient vs analytic	max grad. diff.	0.00e+00	–
Real data identity (lentil/DTF, $r = 0.868$)	abs. diff. (identity)	0.00e+00	–

2 Hyper-parameter search spaces and selected configurations

To keep the headline comparison fair, the architecture and optimizer are tuned once per (dataset, model) with a neutral MSE objective and then frozen across every loss. Table S2 lists the search spaces; Table S3 lists the configuration selected for each dataset.

Table S2: Hyper-parameter search spaces. Architecture and optimizer settings are tuned once per (dataset, model) with a neutral MSE objective, selected by validation Pearson, then frozen and reused across every loss so that only the loss changes in the headline comparison. Classical models tune their regularization analogously; GBLUP is tuning-free. Reproduced from `ccgp/hpo.py`.

Component	Hyper-parameter	Search values
NN: mlp	hidden_dims	(64,), (128, 32), (256, 64), (256, 128, 32), (512, 128)
NN: mlp	dropout	0.1, 0.2, 0.3, 0.5
NN: cnn	channels	(8, 16), (16, 32), (32, 64)
NN: cnn	kernel_size	7, 11, 21
NN: cnn	first_stride	4, 8, 16
NN: cnn	fc_dim	32, 64, 128
NN: cnn	dropout	0.1, 0.2, 0.3
NN: transformer	n_tokens	32, 64, 128
NN: transformer	d_model	32, 64
NN: transformer	n_heads	2, 4
NN: transformer	n_layers	1, 2
NN: transformer	dropout	0.1, 0.2

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Table S2: Hyper-parameter search spaces. Architecture and optimizer settings are tuned once per (dataset, model) with a neutral MSE objective, selected by validation Pearson, then frozen and reused across every loss so that only the loss changes in the headline comparison. Classical models tune their regularization analogously; GBLUP is tuning-free. Reproduced from `ccgp/hpo.py`.

Component	Hyper-parameter	Search values
NN: common (optimizer)	lr	0.003, 0.001, 0.0003
NN: common (optimizer)	weight_decay	0.001, 0.0001, 1e-05
NN: common (optimizer)	batch_size	64, 128, 256
Classical: gblup	–	no tuning
Classical: ridge	alpha	1.0, 10.0, 100.0, 1000.0, 10000.0
Classical: elasticnet	alpha	0.001, 0.01, 0.1, 1.0
Classical: elasticnet	l1_ratio	0.1, 0.5, 0.9
Classical: rf	n_estimators	300
Classical: rf	max_depth	None, 8, 16
Classical: rf	max_features	sqrt, 0.1
Classical: xgboost	max_depth	3, 4, 6
Classical: xgboost	learning_rate	0.03, 0.05, 0.1
Classical: xgboost	n_estimators	300, 500
Classical: xgboost	subsample	0.7, 0.9
Classical: xgboost	colsample_bytree	0.3, 0.5

Table S3: Selected (frozen) hyper-parameters per dataset, from `results/hpo_full.json`. Architecture and optimizer (learning rate, weight decay, batch size) for each neural model, plus the tuned ridge penalty α (shown once per dataset on the MLP row). These are reused unchanged across all losses.

Dataset	Model	Architecture	Optimizer	Ridge α
easygese:barley	mlp	hidden=(256, 128, 32), drop=0.5	lr=0.001, wd=0.0001, bs=256	1000.0
easygese:barley	cnn	ch=(16, 32), k=21, stride=16, fc=64, drop=0.3	lr=0.0003, wd=1e-05, bs=128	
easygese:bean	mlp	hidden=(512, 128), drop=0.3	lr=0.001, wd=0.001, bs=64	10000.0
easygese:bean	cnn	ch=(32, 64), k=11, stride=8, fc=32, drop=0.1	lr=0.003, wd=0.001, bs=64	
easygese:lentil	mlp	hidden=(512, 128), drop=0.1	lr=0.003, wd=1e-05, bs=64	10000.0
easygese:lentil	cnn	ch=(32, 64), k=11, stride=8, fc=32, drop=0.1	lr=0.003, wd=0.001, bs=64	
easygese:lentil	transformer	tok=64, d=64, heads=4, layers=2, drop=0.1	lr=0.0003, wd=1e-05, bs=64	
easygese:maize	mlp	hidden=(256, 128, 32), drop=0.5	lr=0.001, wd=0.0001, bs=256	10000.0
easygese:maize	cnn	ch=(16, 32), k=21, stride=8, fc=32, drop=0.3	lr=0.0003, wd=1e-05, bs=64	

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Table S3: Selected (frozen) hyper-parameters per dataset, from `results/hpo_full.json`. Architecture and optimizer (learning rate, weight decay, batch size) for each neural model, plus the tuned ridge penalty α (shown once per dataset on the MLP row). These are reused unchanged across all losses.

Dataset	Model	Architecture	Optimizer	Ridge α
easygese:maize	transformer	tok=32, d=64, heads=4, layers=1, drop=0.1	lr=0.0003, wd=1e-05, bs=256	
easygese:oyster	mlp	hidden=(256, 128, 32), drop=0.3	lr=0.001, wd=0.0001, bs=256	10000.0
easygese:oyster	cnn	ch=(8, 16), k=21, stride=4, fc=64, drop=0.1	lr=0.003, wd=0.0001, bs=128	
easygese:pig	mlp	hidden=(64,), drop=0.1	lr=0.003, wd=0.001, bs=256	10000.0
easygese:pig	cnn	ch=(8, 16), k=21, stride=8, fc=128, drop=0.2	lr=0.001, wd=1e-05, bs=256	
easygese:pine	mlp	hidden=(128, 32), drop=0.5	lr=0.0003, wd=0.001, bs=128	10000.0
easygese:pine	cnn	ch=(16, 32), k=21, stride=8, fc=32, drop=0.3	lr=0.0003, wd=1e-05, bs=64	
easygese:rice	mlp	hidden=(512, 128), drop=0.3	lr=0.001, wd=0.001, bs=64	10000.0
easygese:rice	cnn	ch=(32, 64), k=11, stride=8, fc=32, drop=0.1	lr=0.003, wd=0.001, bs=64	
easygese:rice	transformer	tok=32, d=64, heads=4, layers=2, drop=0.2	lr=0.001, wd=1e-05, bs=256	
easygese:soybean	mlp	hidden=(64,), drop=0.1	lr=0.003, wd=0.001, bs=256	10000.0
easygese:soybean	cnn	ch=(16, 32), k=21, stride=16, fc=64, drop=0.3	lr=0.0003, wd=1e-05, bs=128	
easygese:soybean	transformer	tok=128, d=64, heads=4, layers=1, drop=0.1	lr=0.003, wd=0.001, bs=64	
easygese:wheatG	mlp	hidden=(256, 64), drop=0.1	lr=0.003, wd=0.0001, bs=128	10000.0
easygese:wheatG	cnn	ch=(16, 32), k=21, stride=16, fc=64, drop=0.3	lr=0.0003, wd=1e-05, bs=128	
wheat:_	mlp	hidden=(512, 128), drop=0.1	lr=0.003, wd=1e-05, bs=64	1000.0
wheat:_	cnn	ch=(16, 32), k=21, stride=16, fc=64, drop=0.3	lr=0.0003, wd=1e-05, bs=128	
wheat:_	transformer	tok=64, d=64, heads=4, layers=1, drop=0.2	lr=0.0003, wd=1e-05, bs=64	
soynam:yield	mlp	hidden=(128, 32), drop=0.5	lr=0.0003, wd=0.001, bs=128	10000.0
soynam:yield	cnn	ch=(16, 32), k=7, stride=4, fc=32, drop=0.1	lr=0.0003, wd=0.0001, bs=128	

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Table S3: Selected (frozen) hyper-parameters per dataset, from `results/hpo_full.json`. Architecture and optimizer (learning rate, weight decay, batch size) for each neural model, plus the tuned ridge penalty α (shown once per dataset on the MLP row). These are reused unchanged across all losses.

Dataset	Model	Architecture	Optimizer	Ridge α
soynam:yield	transformer	tok=64, d=64, heads=4, layers=2, drop=0.2	lr=0.0003, wd=1e-05, bs=128	
soynam:height	mlp	hidden=(128, 32), drop=0.5	lr=0.0003, wd=0.001, bs=128	1000.0
soynam:height	cnn	ch=(32, 64), k=11, stride=8, fc=32, drop=0.1	lr=0.003, wd=0.001, bs=64	
soynam:height	transformer	tok=64, d=64, heads=4, layers=2, drop=0.1	lr=0.0003, wd=1e-05, bs=64	
soynam:protein	mlp	hidden=(128, 32), drop=0.5	lr=0.0003, wd=0.001, bs=128	1000.0
soynam:protein	cnn	ch=(16, 32), k=21, stride=16, fc=64, drop=0.3	lr=0.0003, wd=1e-05, bs=128	
soynam:protein	transformer	tok=32, d=64, heads=4, layers=2, drop=0.2	lr=0.001, wd=1e-05, bs=256	
soynam:oil	mlp	hidden=(128, 32), drop=0.5	lr=0.0003, wd=0.001, bs=128	1000.0
soynam:oil	cnn	ch=(32, 64), k=11, stride=8, fc=32, drop=0.1	lr=0.003, wd=0.001, bs=64	
soynam:oil	transformer	tok=64, d=64, heads=4, layers=1, drop=0.2	lr=0.0003, wd=1e-05, bs=64	

3 Full Δ-versus-MSE tables

Table S4 gives the complete Δ-versus-MSE results for the affine-calibrated neural network across all predictive metrics (Pearson, Spearman, RMSE, MAE, R^2) and all selection metrics (overlap, NDCG, selection differential, relative efficiency, mean of selected) at every selection size $k \in \{5, 10, 15, 20\}$, both pooled across architectures and broken down per architecture.

Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
pearson	pearson	pooled	251	0.0384	[0.0282, 0.0497]	$< 10^{-4}$
pearson	pearson	mlp	101	0.0027	[-0.0004, 0.0055]	0.0462
pearson	pearson	cnn	101	0.0440	[0.0263, 0.0658]	0.0179
pearson	pearson	transformer	49	0.1006	[0.0706, 0.1310]	$< 10^{-4}$
pearson	hybrid	pooled	251	0.0412	[0.0309, 0.0524]	$< 10^{-4}$
pearson	hybrid	mlp	101	0.0044	[0.0014, 0.0077]	0.0018
pearson	hybrid	cnn	101	0.0508	[0.0331, 0.0733]	0.0018
pearson	hybrid	transformer	49	0.0975	[0.0657, 0.1273]	$< 10^{-4}$

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
pearson	ccc	pooled	251	0.0438	[0.0337, 0.0555]	$< 10^{-4}$
pearson	ccc	mlp	101	0.0004	[-0.0038, 0.0036]	0.1749
pearson	ccc	cnn	101	0.0554	[0.0395, 0.0748]	$< 10^{-4}$
pearson	ccc	transformer	49	0.1094	[0.0757, 0.1430]	$< 10^{-4}$
spearman	pearson	pooled	251	0.0370	[0.0278, 0.0475]	$< 10^{-4}$
spearman	pearson	mlp	101	0.0032	[0.0002, 0.0062]	0.0236
spearman	pearson	cnn	101	0.0428	[0.0253, 0.0640]	0.0167
spearman	pearson	transformer	49	0.0950	[0.0713, 0.1216]	$< 10^{-4}$
spearman	hybrid	pooled	251	0.0408	[0.0314, 0.0514]	$< 10^{-4}$
spearman	hybrid	mlp	101	0.0050	[0.0020, 0.0087]	0.0029
spearman	hybrid	cnn	101	0.0504	[0.0332, 0.0720]	0.0008
spearman	hybrid	transformer	49	0.0946	[0.0705, 0.1203]	$< 10^{-4}$
spearman	ccc	pooled	251	0.0428	[0.0336, 0.0540]	$< 10^{-4}$
spearman	ccc	mlp	101	0.0008	[-0.0020, 0.0039]	0.3152
spearman	ccc	cnn	101	0.0553	[0.0395, 0.0735]	$< 10^{-4}$
spearman	ccc	transformer	49	0.1036	[0.0760, 0.1332]	$< 10^{-4}$
rmse	pearson	pooled	251	-0.0099	[-0.0549, 0.0786]	0.0049
rmse	pearson	mlp	101	-0.0456	[-0.2640, -0.0050]	0.0740
rmse	pearson	cnn	101	0.0646	[-0.0192, 0.3402]	0.9116
rmse	pearson	transformer	49	-0.0900	[-0.1534, -0.0505]	$< 10^{-4}$
rmse	hybrid	pooled	251	0.0063	[-0.0415, 0.0969]	0.0002
rmse	hybrid	mlp	101	-0.0061	[-0.0389, 0.0144]	0.1718
rmse	hybrid	cnn	101	0.0796	[-0.0236, 0.3263]	0.9116
rmse	hybrid	transformer	49	-0.1191	[-0.2475, -0.0609]	$< 10^{-4}$
rmse	ccc	pooled	251	-0.0565	[-0.1331, -0.0270]	$< 10^{-4}$
rmse	ccc	mlp	101	-0.0394	[-0.2504, 0.0192]	0.9116
rmse	ccc	cnn	101	-0.0382	[-0.0698, -0.0033]	0.0007
rmse	ccc	transformer	49	-0.1294	[-0.2185, -0.0744]	$< 10^{-4}$
mae	pearson	pooled	251	-0.0148	[-0.0436, 0.0502]	0.0010
mae	pearson	mlp	101	-0.0274	[-0.1089, -0.0085]	0.0747
mae	pearson	cnn	101	0.0299	[-0.0322, 0.2055]	1.0000
mae	pearson	transformer	49	-0.0809	[-0.1398, -0.0353]	$< 10^{-4}$
mae	hybrid	pooled	251	0.0002	[-0.0408, 0.0801]	0.0004
mae	hybrid	mlp	101	-0.0029	[-0.0155, 0.0133]	0.4592
mae	hybrid	cnn	101	0.0540	[-0.0342, 0.2740]	1.0000
mae	hybrid	transformer	49	-0.1044	[-0.2267, -0.0481]	$< 10^{-4}$
mae	ccc	pooled	251	-0.0455	[-0.0843, -0.0224]	$< 10^{-4}$
mae	ccc	mlp	101	-0.0167	[-0.1286, 0.0162]	1.0000
mae	ccc	cnn	101	-0.0409	[-0.0772, -0.0088]	0.0023
mae	ccc	transformer	49	-0.1143	[-0.1965, -0.0666]	$< 10^{-4}$
r2	pearson	pooled	251	0.0147	[0.0053, 0.0222]	0.0017
r2	pearson	mlp	101	0.0008	[-0.0026, 0.0039]	0.7571
r2	pearson	cnn	101	0.0049	[-0.0160, 0.0196]	0.8270
r2	pearson	transformer	49	0.0637	[0.0470, 0.0826]	$< 10^{-4}$
r2	hybrid	pooled	251	0.0168	[0.0064, 0.0242]	$< 10^{-4}$

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
r2	hybrid	mlp	101	0.0028	[0.0003, 0.0055]	0.1838
r2	hybrid	cnn	101	0.0097	[-0.0148, 0.0246]	0.7571
r2	hybrid	transformer	49	0.0603	[0.0439, 0.0805]	$< 10^{-4}$
r2	ccc	pooled	251	0.0248	[0.0187, 0.0312]	$< 10^{-4}$
r2	ccc	mlp	101	0.0024	[-0.0007, 0.0058]	0.7571
r2	ccc	cnn	101	0.0247	[0.0165, 0.0343]	$< 10^{-4}$
r2	ccc	transformer	49	0.0710	[0.0518, 0.0918]	$< 10^{-4}$
overlap@5	pearson	pooled	251	0.0108	[0.0038, 0.0178]	0.0056
overlap@5	pearson	mlp	101	0.0029	[-0.0060, 0.0112]	0.6742
overlap@5	pearson	cnn	101	0.0166	[0.0041, 0.0302]	0.2746
overlap@5	pearson	transformer	49	0.0152	[0.0005, 0.0303]	0.2746
overlap@5	hybrid	pooled	251	0.0134	[0.0063, 0.0203]	0.0006
overlap@5	hybrid	mlp	101	0.0067	[-0.0028, 0.0157]	0.5696
overlap@5	hybrid	cnn	101	0.0190	[0.0072, 0.0330]	0.0513
overlap@5	hybrid	transformer	49	0.0156	[0.0003, 0.0289]	0.1208
overlap@5	ccc	pooled	251	0.0121	[0.0056, 0.0186]	0.0053
overlap@5	ccc	mlp	101	0.0022	[-0.0064, 0.0108]	0.7757
overlap@5	ccc	cnn	101	0.0220	[0.0111, 0.0338]	0.0040
overlap@5	ccc	transformer	49	0.0118	[-0.0029, 0.0267]	0.6742
overlap@10	pearson	pooled	251	0.0142	[0.0079, 0.0206]	0.0006
overlap@10	pearson	mlp	101	0.0015	[-0.0054, 0.0085]	1.0000
overlap@10	pearson	cnn	101	0.0187	[0.0076, 0.0307]	0.1245
overlap@10	pearson	transformer	49	0.0312	[0.0159, 0.0475]	0.0017
overlap@10	hybrid	pooled	251	0.0178	[0.0120, 0.0240]	$< 10^{-4}$
overlap@10	hybrid	mlp	101	0.0049	[-0.0016, 0.0115]	0.5671
overlap@10	hybrid	cnn	101	0.0242	[0.0134, 0.0361]	0.0039
overlap@10	hybrid	transformer	49	0.0313	[0.0170, 0.0431]	0.0001
overlap@10	ccc	pooled	251	0.0170	[0.0113, 0.0233]	$< 10^{-4}$
overlap@10	ccc	mlp	101	0.0009	[-0.0058, 0.0083]	1.0000
overlap@10	ccc	cnn	101	0.0254	[0.0151, 0.0363]	0.0002
overlap@10	ccc	transformer	49	0.0328	[0.0185, 0.0460]	0.0005
overlap@15	pearson	pooled	251	0.0159	[0.0098, 0.0221]	0.0003
overlap@15	pearson	mlp	101	-0.0000	[-0.0058, 0.0063]	1.0000
overlap@15	pearson	cnn	101	0.0244	[0.0140, 0.0359]	0.0043
overlap@15	pearson	transformer	49	0.0311	[0.0180, 0.0471]	0.0043
overlap@15	hybrid	pooled	251	0.0193	[0.0133, 0.0256]	$< 10^{-4}$
overlap@15	hybrid	mlp	101	0.0017	[-0.0035, 0.0075]	1.0000
overlap@15	hybrid	cnn	101	0.0262	[0.0153, 0.0388]	0.0041
overlap@15	hybrid	transformer	49	0.0412	[0.0272, 0.0585]	$< 10^{-4}$
overlap@15	ccc	pooled	251	0.0190	[0.0133, 0.0251]	$< 10^{-4}$
overlap@15	ccc	mlp	101	-0.0032	[-0.0094, 0.0027]	1.0000
overlap@15	ccc	cnn	101	0.0282	[0.0188, 0.0381]	$< 10^{-4}$
overlap@15	ccc	transformer	49	0.0457	[0.0317, 0.0613]	$< 10^{-4}$
overlap@20	pearson	pooled	251	0.0171	[0.0116, 0.0228]	$< 10^{-4}$
overlap@20	pearson	mlp	101	0.0027	[-0.0019, 0.0074]	0.8606

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
overlap@20	pearson	cnn	101	0.0208	[0.0114, 0.0321]	0.0143
overlap@20	pearson	transformer	49	0.0392	[0.0251, 0.0559]	0.0002
overlap@20	hybrid	pooled	251	0.0186	[0.0133, 0.0243]	$< 10^{-4}$
overlap@20	hybrid	mlp	101	0.0037	[-0.0005, 0.0081]	0.3294
overlap@20	hybrid	cnn	101	0.0249	[0.0149, 0.0363]	0.0021
overlap@20	hybrid	transformer	49	0.0363	[0.0240, 0.0509]	0.0001
overlap@20	ccc	pooled	251	0.0203	[0.0151, 0.0261]	$< 10^{-4}$
overlap@20	ccc	mlp	101	0.0000	[-0.0044, 0.0049]	0.8606
overlap@20	ccc	cnn	101	0.0277	[0.0189, 0.0375]	$< 10^{-4}$
overlap@20	ccc	transformer	49	0.0467	[0.0320, 0.0634]	$< 10^{-4}$
ndcg@5	pearson	pooled	251	0.0155	[0.0100, 0.0214]	$< 10^{-4}$
ndcg@5	pearson	mlp	101	0.0040	[-0.0019, 0.0094]	0.1183
ndcg@5	pearson	cnn	101	0.0228	[0.0132, 0.0339]	0.0031
ndcg@5	pearson	transformer	49	0.0242	[0.0102, 0.0422]	0.0065
ndcg@5	hybrid	pooled	251	0.0176	[0.0123, 0.0230]	$< 10^{-4}$
ndcg@5	hybrid	mlp	101	0.0078	[0.0031, 0.0134]	0.1183
ndcg@5	hybrid	cnn	101	0.0251	[0.0158, 0.0361]	0.0011
ndcg@5	hybrid	transformer	49	0.0224	[0.0076, 0.0361]	0.0065
ndcg@5	ccc	pooled	251	0.0166	[0.0115, 0.0222]	$< 10^{-4}$
ndcg@5	ccc	mlp	101	0.0036	[-0.0016, 0.0097]	0.6856
ndcg@5	ccc	cnn	101	0.0262	[0.0173, 0.0363]	$< 10^{-4}$
ndcg@5	ccc	transformer	49	0.0239	[0.0108, 0.0360]	0.0009
ndcg@10	pearson	pooled	251	0.0161	[0.0113, 0.0215]	$< 10^{-4}$
ndcg@10	pearson	mlp	101	0.0026	[-0.0014, 0.0064]	0.2821
ndcg@10	pearson	cnn	101	0.0234	[0.0149, 0.0334]	0.0020
ndcg@10	pearson	transformer	49	0.0291	[0.0169, 0.0466]	$< 10^{-4}$
ndcg@10	hybrid	pooled	251	0.0173	[0.0127, 0.0221]	$< 10^{-4}$
ndcg@10	hybrid	mlp	101	0.0050	[0.0016, 0.0090]	0.1339
ndcg@10	hybrid	cnn	101	0.0248	[0.0167, 0.0347]	0.0007
ndcg@10	hybrid	transformer	49	0.0270	[0.0142, 0.0394]	$< 10^{-4}$
ndcg@10	ccc	pooled	251	0.0173	[0.0126, 0.0222]	$< 10^{-4}$
ndcg@10	ccc	mlp	101	0.0011	[-0.0030, 0.0053]	0.5013
ndcg@10	ccc	cnn	101	0.0271	[0.0191, 0.0355]	$< 10^{-4}$
ndcg@10	ccc	transformer	49	0.0304	[0.0170, 0.0424]	$< 10^{-4}$
ndcg@15	pearson	pooled	251	0.0150	[0.0103, 0.0199]	$< 10^{-4}$
ndcg@15	pearson	mlp	101	0.0012	[-0.0028, 0.0044]	0.2116
ndcg@15	pearson	cnn	101	0.0221	[0.0144, 0.0315]	0.0007
ndcg@15	pearson	transformer	49	0.0288	[0.0166, 0.0443]	$< 10^{-4}$
ndcg@15	hybrid	pooled	251	0.0166	[0.0123, 0.0212]	$< 10^{-4}$
ndcg@15	hybrid	mlp	101	0.0038	[0.0010, 0.0071]	0.2116
ndcg@15	hybrid	cnn	101	0.0236	[0.0161, 0.0332]	0.0002
ndcg@15	hybrid	transformer	49	0.0286	[0.0159, 0.0406]	$< 10^{-4}$
ndcg@15	ccc	pooled	251	0.0174	[0.0130, 0.0224]	$< 10^{-4}$
ndcg@15	ccc	mlp	101	0.0005	[-0.0030, 0.0040]	0.7361
ndcg@15	ccc	cnn	101	0.0263	[0.0190, 0.0347]	$< 10^{-4}$

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
ndcg@15	ccc	transformer	49	0.0341	[0.0209, 0.0464]	$< 10^{-4}$
ndcg@20	pearson	pooled	251	0.0150	[0.0107, 0.0196]	$< 10^{-4}$
ndcg@20	pearson	mlp	101	0.0015	[-0.0021, 0.0044]	0.2506
ndcg@20	pearson	cnn	101	0.0207	[0.0134, 0.0292]	0.0004
ndcg@20	pearson	transformer	49	0.0314	[0.0208, 0.0454]	$< 10^{-4}$
ndcg@20	hybrid	pooled	251	0.0160	[0.0120, 0.0204]	$< 10^{-4}$
ndcg@20	hybrid	mlp	101	0.0033	[0.0010, 0.0063]	0.1843
ndcg@20	hybrid	cnn	101	0.0227	[0.0155, 0.0315]	$< 10^{-4}$
ndcg@20	hybrid	transformer	49	0.0283	[0.0159, 0.0391]	$< 10^{-4}$
ndcg@20	ccc	pooled	251	0.0171	[0.0129, 0.0217]	$< 10^{-4}$
ndcg@20	ccc	mlp	101	0.0005	[-0.0028, 0.0034]	0.4842
ndcg@20	ccc	cnn	101	0.0250	[0.0178, 0.0329]	$< 10^{-4}$
ndcg@20	ccc	transformer	49	0.0349	[0.0236, 0.0468]	$< 10^{-4}$
seldiff@5	pearson	pooled	251	0.4063	[0.1970, 1.0544]	$< 10^{-4}$
seldiff@5	pearson	mlp	101	0.4748	[0.0421, 2.4713]	0.1784
seldiff@5	pearson	cnn	101	0.4100	[0.1483, 0.7808]	0.0358
seldiff@5	pearson	transformer	49	0.2575	[-0.0001, 0.6040]	0.0045
seldiff@5	hybrid	pooled	251	0.2188	[-0.0222, 0.4768]	$< 10^{-4}$
seldiff@5	hybrid	mlp	101	0.0387	[-0.4879, 0.3128]	0.1784
seldiff@5	hybrid	cnn	101	0.4310	[0.0383, 0.9921]	0.0137
seldiff@5	hybrid	transformer	49	0.1525	[-0.1971, 0.4405]	0.0130
seldiff@5	ccc	pooled	251	0.5222	[0.2904, 1.0224]	$< 10^{-4}$
seldiff@5	ccc	mlp	101	0.3303	[0.0129, 1.7296]	0.7349
seldiff@5	ccc	cnn	101	0.7825	[0.4196, 1.7657]	0.0001
seldiff@5	ccc	transformer	49	0.3811	[0.0566, 0.8431]	0.0008
seldiff@10	pearson	pooled	251	0.2516	[0.0188, 0.6118]	$< 10^{-4}$
seldiff@10	pearson	mlp	101	0.3024	[-0.0098, 1.7537]	0.1562
seldiff@10	pearson	cnn	101	0.1380	[-0.6376, 0.4274]	0.0780
seldiff@10	pearson	transformer	49	0.3812	[0.1810, 0.8803]	$< 10^{-4}$
seldiff@10	hybrid	pooled	251	0.1657	[-0.1976, 0.3924]	$< 10^{-4}$
seldiff@10	hybrid	mlp	101	0.1123	[-0.0681, 0.4429]	0.1562
seldiff@10	hybrid	cnn	101	0.1247	[-0.8140, 0.6074]	0.0431
seldiff@10	hybrid	transformer	49	0.3605	[0.1882, 0.7226]	$< 10^{-4}$
seldiff@10	ccc	pooled	251	0.3973	[0.1981, 0.7863]	$< 10^{-4}$
seldiff@10	ccc	mlp	101	0.3211	[0.0423, 1.7907]	0.1562
seldiff@10	ccc	cnn	101	0.4024	[0.0821, 0.8599]	$< 10^{-4}$
seldiff@10	ccc	transformer	49	0.5437	[0.2253, 1.3723]	0.0001
seldiff@15	pearson	pooled	251	0.1663	[-0.0286, 0.4746]	0.0001
seldiff@15	pearson	mlp	101	0.1858	[-0.0927, 1.3171]	1.0000
seldiff@15	pearson	cnn	101	0.1048	[-0.4690, 0.3787]	0.0958
seldiff@15	pearson	transformer	49	0.2528	[0.1373, 0.5004]	$< 10^{-4}$
seldiff@15	hybrid	pooled	251	0.1360	[-0.1040, 0.3379]	$< 10^{-4}$
seldiff@15	hybrid	mlp	101	0.0257	[-0.2179, 0.1976]	1.0000
seldiff@15	hybrid	cnn	101	0.1210	[-0.4690, 0.5430]	0.0572
seldiff@15	hybrid	transformer	49	0.3945	[0.2147, 0.7606]	$< 10^{-4}$

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
seldiff@15	ccc	pooled	251	0.3293	[0.1229, 0.7301]	$< 10^{-4}$
seldiff@15	ccc	mlp	101	0.2122	[-0.0855, 1.4049]	1.0000
seldiff@15	ccc	cnn	101	0.3770	[0.0162, 0.9044]	0.0001
seldiff@15	ccc	transformer	49	0.4721	[0.2438, 0.9264]	$< 10^{-4}$
seldiff@20	pearson	pooled	251	0.2395	[0.1242, 0.4987]	$< 10^{-4}$
seldiff@20	pearson	mlp	101	0.2096	[0.0074, 1.1582]	1.0000
seldiff@20	pearson	cnn	101	0.2475	[0.0507, 0.4478]	0.0457
seldiff@20	pearson	transformer	49	0.2843	[0.1645, 0.5042]	$< 10^{-4}$
seldiff@20	hybrid	pooled	251	0.2263	[0.0982, 0.4417]	$< 10^{-4}$
seldiff@20	hybrid	mlp	101	0.0806	[-0.0483, 0.2804]	1.0000
seldiff@20	hybrid	cnn	101	0.2562	[0.0236, 0.7847]	0.0457
seldiff@20	hybrid	transformer	49	0.4648	[0.2223, 1.0229]	$< 10^{-4}$
seldiff@20	ccc	pooled	251	0.3070	[0.1520, 0.6326]	$< 10^{-4}$
seldiff@20	ccc	mlp	101	0.1742	[-0.0286, 1.1018]	1.0000
seldiff@20	ccc	cnn	101	0.3850	[0.1469, 1.0167]	0.0001
seldiff@20	ccc	transformer	49	0.4199	[0.2220, 0.8069]	$< 10^{-4}$
releff@5	pearson	pooled	251	0.0314	[0.0163, 0.0441]	$< 10^{-4}$
releff@5	pearson	mlp	101	0.0068	[-0.0045, 0.0193]	0.3287
releff@5	pearson	cnn	101	0.0435	[0.0109, 0.0698]	0.0052
releff@5	pearson	transformer	49	0.0569	[0.0317, 0.0853]	0.0019
releff@5	hybrid	pooled	251	0.0359	[0.0228, 0.0488]	$< 10^{-4}$
releff@5	hybrid	mlp	101	0.0150	[0.0023, 0.0282]	0.2197
releff@5	hybrid	cnn	101	0.0503	[0.0236, 0.0777]	0.0030
releff@5	hybrid	transformer	49	0.0492	[0.0188, 0.0762]	0.0052
releff@5	ccc	pooled	251	0.0334	[0.0204, 0.0450]	$< 10^{-4}$
releff@5	ccc	mlp	101	0.0075	[-0.0038, 0.0210]	0.5775
releff@5	ccc	cnn	101	0.0526	[0.0271, 0.0732]	$< 10^{-4}$
releff@5	ccc	transformer	49	0.0470	[0.0148, 0.0729]	0.0013
releff@10	pearson	pooled	251	0.0383	[0.0275, 0.0505]	$< 10^{-4}$
releff@10	pearson	mlp	101	0.0062	[-0.0030, 0.0151]	0.1574
releff@10	pearson	cnn	101	0.0532	[0.0328, 0.0773]	0.0042
releff@10	pearson	transformer	49	0.0740	[0.0492, 0.1038]	$< 10^{-4}$
releff@10	hybrid	pooled	251	0.0392	[0.0286, 0.0509]	$< 10^{-4}$
releff@10	hybrid	mlp	101	0.0079	[-0.0012, 0.0163]	0.2092
releff@10	hybrid	cnn	101	0.0560	[0.0358, 0.0816]	0.0044
releff@10	hybrid	transformer	49	0.0692	[0.0460, 0.0934]	$< 10^{-4}$
releff@10	ccc	pooled	251	0.0396	[0.0285, 0.0514]	$< 10^{-4}$
releff@10	ccc	mlp	101	-0.0006	[-0.0119, 0.0091]	0.6278
releff@10	ccc	cnn	101	0.0622	[0.0430, 0.0829]	$< 10^{-4}$
releff@10	ccc	transformer	49	0.0759	[0.0478, 0.1055]	$< 10^{-4}$
releff@15	pearson	pooled	251	0.0358	[0.0248, 0.0476]	$< 10^{-4}$
releff@15	pearson	mlp	101	0.0019	[-0.0055, 0.0100]	1.0000
releff@15	pearson	cnn	101	0.0513	[0.0321, 0.0752]	0.0036
releff@15	pearson	transformer	49	0.0735	[0.0465, 0.1056]	$< 10^{-4}$
releff@15	hybrid	pooled	251	0.0401	[0.0291, 0.0523]	$< 10^{-4}$

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
releff@15	hybrid	mlp	101	0.0059	[-0.0009, 0.0137]	1.0000
releff@15	hybrid	cnn	101	0.0550	[0.0345, 0.0802]	0.0011
releff@15	hybrid	transformer	49	0.0801	[0.0541, 0.1105]	$< 10^{-4}$
releff@15	ccc	pooled	251	0.0444	[0.0334, 0.0568]	$< 10^{-4}$
releff@15	ccc	mlp	101	0.0004	[-0.0064, 0.0076]	1.0000
releff@15	ccc	cnn	101	0.0646	[0.0454, 0.0866]	$< 10^{-4}$
releff@15	ccc	transformer	49	0.0936	[0.0664, 0.1258]	$< 10^{-4}$
releff@20	pearson	pooled	251	0.0387	[0.0286, 0.0501]	$< 10^{-4}$
releff@20	pearson	mlp	101	0.0026	[-0.0039, 0.0100]	1.0000
releff@20	pearson	cnn	101	0.0486	[0.0308, 0.0701]	0.0007
releff@20	pearson	transformer	49	0.0925	[0.0660, 0.1242]	$< 10^{-4}$
releff@20	hybrid	pooled	251	0.0403	[0.0297, 0.0522]	$< 10^{-4}$
releff@20	hybrid	mlp	101	0.0042	[-0.0024, 0.0115]	1.0000
releff@20	hybrid	cnn	101	0.0548	[0.0360, 0.0791]	0.0003
releff@20	hybrid	transformer	49	0.0848	[0.0571, 0.1148]	$< 10^{-4}$
releff@20	ccc	pooled	251	0.0451	[0.0346, 0.0576]	$< 10^{-4}$
releff@20	ccc	mlp	101	0.0002	[-0.0057, 0.0067]	1.0000
releff@20	ccc	cnn	101	0.0633	[0.0444, 0.0845]	$< 10^{-4}$
releff@20	ccc	transformer	49	0.1000	[0.0738, 0.1313]	$< 10^{-4}$
meansel@5	pearson	pooled	251	0.4063	[0.1970, 1.0544]	$< 10^{-4}$
meansel@5	pearson	mlp	101	0.4748	[0.0421, 2.4713]	0.1828
meansel@5	pearson	cnn	101	0.4100	[0.1483, 0.7808]	0.0338
meansel@5	pearson	transformer	49	0.2575	[-0.0001, 0.6040]	0.0045
meansel@5	hybrid	pooled	251	0.2188	[-0.0222, 0.4768]	$< 10^{-4}$
meansel@5	hybrid	mlp	101	0.0387	[-0.4879, 0.3128]	0.1828
meansel@5	hybrid	cnn	101	0.4310	[0.0383, 0.9921]	0.0137
meansel@5	hybrid	transformer	49	0.1525	[-0.1971, 0.4405]	0.0130
meansel@5	ccc	pooled	251	0.5222	[0.2904, 1.0224]	$< 10^{-4}$
meansel@5	ccc	mlp	101	0.3303	[0.0129, 1.7296]	0.7349
meansel@5	ccc	cnn	101	0.7825	[0.4196, 1.7657]	0.0001
meansel@5	ccc	transformer	49	0.3811	[0.0566, 0.8431]	0.0008
meansel@10	pearson	pooled	251	0.2516	[0.0188, 0.6118]	$< 10^{-4}$
meansel@10	pearson	mlp	101	0.3024	[-0.0098, 1.7537]	0.1562
meansel@10	pearson	cnn	101	0.1380	[-0.6376, 0.4274]	0.0780
meansel@10	pearson	transformer	49	0.3812	[0.1810, 0.8803]	$< 10^{-4}$
meansel@10	hybrid	pooled	251	0.1657	[-0.1976, 0.3924]	$< 10^{-4}$
meansel@10	hybrid	mlp	101	0.1123	[-0.0681, 0.4429]	0.1562
meansel@10	hybrid	cnn	101	0.1247	[-0.8140, 0.6074]	0.0431
meansel@10	hybrid	transformer	49	0.3605	[0.1882, 0.7226]	$< 10^{-4}$
meansel@10	ccc	pooled	251	0.3973	[0.1981, 0.7863]	$< 10^{-4}$
meansel@10	ccc	mlp	101	0.3211	[0.0423, 1.7907]	0.1562
meansel@10	ccc	cnn	101	0.4024	[0.0821, 0.8599]	$< 10^{-4}$
meansel@10	ccc	transformer	49	0.5437	[0.2253, 1.3723]	0.0001
meansel@15	pearson	pooled	251	0.1663	[-0.0286, 0.4746]	0.0001
meansel@15	pearson	mlp	101	0.1858	[-0.0927, 1.3171]	1.0000

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Table S4: Delta-versus-MSE for the affine-calibrated neural network, for every predictive and selection metric, pooled across architectures and broken down per architecture (mlp/cnn/transformer). Mean change relative to the MSE baseline with BCa 95% bootstrap confidence interval and Holm-corrected paired-Wilcoxon p-value; n is the number of (dataset, trait) cells. Positive values favour the correlation-consistent loss for pearson/spearman/R^2 and the selection metrics, negative values for rmse/mae.

Metric	Loss	Architecture	n	Mean Δ	95% CI	p_{Holm}
meansel@15	pearson	cnn	101	0.1048	[-0.4690, 0.3787]	0.0958
meansel@15	pearson	transformer	49	0.2528	[0.1373, 0.5004]	$< 10^{-4}$
meansel@15	hybrid	pooled	251	0.1360	[-0.1040, 0.3379]	$< 10^{-4}$
meansel@15	hybrid	mlp	101	0.0257	[-0.2179, 0.1976]	1.0000
meansel@15	hybrid	cnn	101	0.1210	[-0.4690, 0.5430]	0.0572
meansel@15	hybrid	transformer	49	0.3945	[0.2147, 0.7606]	$< 10^{-4}$
meansel@15	ccc	pooled	251	0.3293	[0.1229, 0.7301]	$< 10^{-4}$
meansel@15	ccc	mlp	101	0.2122	[-0.0855, 1.4049]	1.0000
meansel@15	ccc	cnn	101	0.3770	[0.0162, 0.9044]	0.0001
meansel@15	ccc	transformer	49	0.4721	[0.2438, 0.9264]	$< 10^{-4}$
meansel@20	pearson	pooled	251	0.2395	[0.1242, 0.4987]	$< 10^{-4}$
meansel@20	pearson	mlp	101	0.2096	[0.0074, 1.1582]	1.0000
meansel@20	pearson	cnn	101	0.2475	[0.0507, 0.4478]	0.0457
meansel@20	pearson	transformer	49	0.2843	[0.1645, 0.5042]	$< 10^{-4}$
meansel@20	hybrid	pooled	251	0.2263	[0.0982, 0.4417]	$< 10^{-4}$
meansel@20	hybrid	mlp	101	0.0806	[-0.0483, 0.2804]	1.0000
meansel@20	hybrid	cnn	101	0.2562	[0.0236, 0.7847]	0.0457
meansel@20	hybrid	transformer	49	0.4648	[0.2223, 1.0229]	$< 10^{-4}$
meansel@20	ccc	pooled	251	0.3070	[0.1520, 0.6326]	$< 10^{-4}$
meansel@20	ccc	mlp	101	0.1742	[-0.0286, 1.1018]	1.0000
meansel@20	ccc	cnn	101	0.3850	[0.1469, 1.0167]	0.0001
meansel@20	ccc	transformer	49	0.4199	[0.2220, 0.8069]	$< 10^{-4}$

4 Confirmatory mixed-effects models

Table S5 reports the linear mixed-model coefficients (lme4) for Pearson r, NDCG@10 and RMSE on the affine-calibrated grid, with the MSE loss and the CNN architecture as references and a random intercept per (dataset, trait).

Table S5: Mixed-effects model coefficients for the affine-calibrated neural-network grid. Estimates, standard errors, t-values and 95% confidence intervals for each fixed effect; the MSE loss and the CNN architecture are the reference levels.

Response: Pearson r.

Term	Estimate	SE	t	95% CI
(Intercept)	0.4894	0.0499	9.80	[0.3915, 0.5872]
lossccc	0.0438	0.0054	8.12	[0.0332, 0.0544]
losshybrid	0.0412	0.0054	7.63	[0.0306, 0.0518]
losspearson	0.0384	0.0054	7.11	[0.0278, 0.0490]
modelmlp	0.0608	0.0043	14.29	[0.0525, 0.0692]
modeltransformer	-0.0197	0.0057	-3.46	[-0.0309, -0.0086]

Response: NDCG@10.

Term	Estimate	SE	t	95% CI
(Intercept)	0.7559	0.0314	24.11	[0.6945, 0.8174]
lossccc	0.0173	0.0027	6.51	[0.0121, 0.0225]
losshybrid	0.0173	0.0027	6.52	[0.0121, 0.0225]
losspearson	0.0161	0.0027	6.09	[0.0109, 0.0213]
modelmlp	0.0239	0.0021	11.46	[0.0198, 0.0280]
modeltransformer	-0.0104	0.0028	-3.71	[-0.0159, -0.0049]

Response: RMSE.

Term	Estimate	SE	t	95% CI
(Intercept)	18.5031	12.4139	1.49	[-5.8278, 42.8340]
lossccc	-0.0565	0.0681	-0.83	[-0.1900, 0.0770]
losshybrid	0.0063	0.0681	0.09	[-0.1272, 0.1398]
losspearson	-0.0099	0.0681	-0.15	[-0.1434, 0.1236]
modelmlp	-0.4347	0.0537	-8.10	[-0.5399, -0.3295]
modeltransformer	-0.1257	0.0720	-1.75	[-0.2667, 0.0154]

5 Mean method ranks

Table S6 ranks every model/loss combination across all trait-datasets for predictive ability and for selection quality.

Table S6: Mean rank of every model/loss method across all trait-datasets (affine-calibrated), lower is better, for predictive ability (Pearson r) and selection quality (NDCG@10). Ranks are computed within each (dataset, trait, scheme, calibration) block and averaged. GBLUP and ridge use no neural loss (`na`).

Method (model/loss)	Mean rank (Pearson)	Mean rank (NDCG@10)
ridge/na	2.85	4.01
gblup/na	3.45	4.32
mlp/hybrid	5.28	4.80
mlp/pearson	5.47	4.93
mlp/ccc	5.62	5.30
mlp/mse	6.14	5.56
cnn/ccc	6.47	7.05
cnn/hybrid	7.22	7.36
cnn/pearson	7.93	7.91
transformer/ccc	8.57	8.92
cnn/mse	8.94	9.18
transformer/pearson	9.67	8.92
transformer/hybrid	9.88	9.20
transformer/mse	12.67	11.78

6 Calibration

Table S7 summarizes the effect of post-hoc calibration on the Pearson-trained network. Figure S3 shows the fold-level raw-versus-affine relationship for the calibration slope, RMSE and intercept.

Table S7: Effect of post-hoc calibration on the Pearson-trained neural network (mlp/cnn/transformer): mean $\pm$ s.d. of each diagnostic across all folds. Affine calibration restores the regression slope towards 1 and reduces RMSE while leaving the rank correlation unchanged; isotonic calibration is monotone and similar.

Calibration	Pearson r	RMSE	Slope	Intercept	Bias
raw	0.531 $\pm$ 0.229	8.682 $\pm$ 33.322	4.253 $\pm$ 20.829	-378.780 $\pm$ 5079.496	-0.755 $\pm$ 13.857
affine	0.533 $\pm$ 0.227	7.065 $\pm$ 26.142	2.269 $\pm$ 66.562	0.104 $\pm$ 161.330	0.044 $\pm$ 5.005
isotonic	0.516 $\pm$ 0.229	7.263 $\pm$ 26.824	0.764 $\pm$ 0.308	9.386 $\pm$ 101.289	0.044 $\pm$ 4.973

7 Selection metrics across selection intensities

Figure S2 shows how the selection benefit varies with the selection size k.

8 Minibatch size (Experiment E)

Table S8 reports test accuracy as a function of minibatch size for the Pearson loss; Figure S5 shows the per-dataset curves. The minibatch estimator of the standardized-MSE loss is consistent, so accuracy is stable across batch sizes.

Table S8: Experiment E: minibatch size and the Pearson loss. Mean $\pm$ s.d. of test Pearson r, RMSE and NDCG@10 (affine-calibrated MLP) over four EasyGeSe datasets as a function of minibatch size (full = full-batch gradient). The minibatch estimator of the standardized-MSE loss is consistent: accuracy is stable across batch sizes.

Loss	Batch size	Pearson r	RMSE	NDCG@10
mse	full	0.5685 $\pm$ 0.2765	7.624 $\pm$ 10.750	0.7709 $\pm$ 0.1835
mse	32	0.5685 $\pm$ 0.2676	7.616 $\pm$ 10.583	0.7730 $\pm$ 0.1740
mse	128	0.5676 $\pm$ 0.2846	7.604 $\pm$ 10.684	0.7740 $\pm$ 0.1782
mse	512	0.5690 $\pm$ 0.2766	7.626 $\pm$ 10.750	0.7714 $\pm$ 0.1838
pearson	full	0.5782 $\pm$ 0.2630	7.560 $\pm$ 10.573	0.7821 $\pm$ 0.1721
pearson	32	0.5774 $\pm$ 0.2615	7.614 $\pm$ 10.696	0.7752 $\pm$ 0.1798
pearson	128	0.5777 $\pm$ 0.2610	7.616 $\pm$ 10.703	0.7754 $\pm$ 0.1797
pearson	512	0.5790 $\pm$ 0.2635	7.554 $\pm$ 10.574	0.7820 $\pm$ 0.1720

9 Difficult splits (Experiment F)

Tables S9 and S10 ask whether the loss benefit persists under random, within-environment, leave-family-out and cross-environment cross-validation, pooled and per architecture.

Table S9: Experiment F: does correlation-consistent training help under harder splits? Delta-versus-MSE (affine-calibrated NN) by cross-validation scheme (random, within-environment, leave-family-out, cross-environment), pooled across architectures. Mean with BCa 95% CI and Holm-corrected paired-Wilcoxon p. The benefit is reliable under random/within-environment splits and attenuates under family/environment extrapolation.

Metric	Loss	Scheme	n	Mean Δ	95% CI	p_{Holm}
pearson	pearson	random	12	0.0158	[0.0031, 0.0422]	0.5757
pearson	pearson	within_env	12	0.0131	[-0.0066, 0.0435]	1.0000
pearson	pearson	family	12	-0.0062	[-0.0172, 0.0038]	1.0000
pearson	pearson	cross_env	36	-0.0019	[-0.0112, 0.0070]	1.0000
pearson	hybrid	random	12	0.0190	[0.0065, 0.0434]	0.2954
pearson	hybrid	within_env	12	0.0188	[-0.0005, 0.0468]	1.0000
pearson	hybrid	family	12	-0.0001	[-0.0115, 0.0097]	1.0000
pearson	hybrid	cross_env	36	-0.0047	[-0.0133, 0.0046]	1.0000
pearson	ccc	random	12	0.0246	[0.0124, 0.0544]	0.0059
pearson	ccc	within_env	12	0.0273	[0.0098, 0.0551]	0.2954
pearson	ccc	family	12	0.0035	[-0.0033, 0.0140]	1.0000
pearson	ccc	cross_env	36	-0.0052	[-0.0157, 0.0053]	1.0000
ndcg@10	pearson	random	12	0.0089	[0.0015, 0.0323]	0.5225
ndcg@10	pearson	within_env	12	0.0073	[-0.0047, 0.0214]	1.0000
ndcg@10	pearson	family	12	0.0026	[-0.0057, 0.0102]	1.0000
ndcg@10	pearson	cross_env	36	-0.0018	[-0.0090, 0.0055]	1.0000
ndcg@10	hybrid	random	12	0.0114	[0.0030, 0.0364]	0.4673
ndcg@10	hybrid	within_env	12	0.0141	[0.0026, 0.0311]	1.0000
ndcg@10	hybrid	family	12	0.0011	[-0.0085, 0.0105]	1.0000
ndcg@10	hybrid	cross_env	36	0.0016	[-0.0046, 0.0080]	1.0000
ndcg@10	ccc	random	12	0.0121	[0.0045, 0.0397]	0.0059
ndcg@10	ccc	within_env	12	0.0128	[0.0003, 0.0333]	1.0000
ndcg@10	ccc	family	12	0.0049	[-0.0023, 0.0146]	1.0000
ndcg@10	ccc	cross_env	36	0.0007	[-0.0067, 0.0079]	1.0000
overlap@10	pearson	random	12	0.0196	[0.0092, 0.0475]	0.0376
overlap@10	pearson	within_env	12	0.0181	[-0.0014, 0.0484]	1.0000
overlap@10	pearson	family	12	-0.0014	[-0.0125, 0.0088]	1.0000
overlap@10	pearson	cross_env	36	-0.0025	[-0.0116, 0.0079]	1.0000
overlap@10	hybrid	random	12	0.0201	[0.0072, 0.0437]	0.3174
overlap@10	hybrid	within_env	12	0.0271	[0.0007, 0.0583]	1.0000
overlap@10	hybrid	family	12	-0.0060	[-0.0215, 0.0112]	1.0000
overlap@10	hybrid	cross_env	36	-0.0016	[-0.0104, 0.0088]	1.0000
overlap@10	ccc	random	12	0.0219	[0.0114, 0.0510]	0.0117
overlap@10	ccc	within_env	12	0.0111	[-0.0111, 0.0403]	1.0000
overlap@10	ccc	family	12	-0.0057	[-0.0139, 0.0028]	1.0000
overlap@10	ccc	cross_env	36	-0.0046	[-0.0144, 0.0058]	1.0000
releff@10	pearson	random	12	0.0278	[0.0082, 0.0854]	0.1772
releff@10	pearson	within_env	12	0.0219	[-0.0107, 0.0637]	1.0000
releff@10	pearson	family	12	0.0025	[-0.0144, 0.0172]	1.0000

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Table S9: Experiment F: does correlation-consistent training help under harder splits? Delta-versus-MSE (affine-calibrated NN) by cross-validation scheme (random, within-environment, leave-family-out, cross-environment), pooled across architectures. Mean with BCa 95% CI and Holm-corrected paired-Wilcoxon p. The benefit is reliable under random/within-environment splits and attenuates under family/environment extrapolation.

Metric	Loss	Scheme	n	Mean Δ	95% CI	p_{Holm}
releff@10	pearson	cross_env	36	0.0012	[-0.0192, 0.0216]	1.0000
releff@10	hybrid	random	12	0.0309	[0.0109, 0.0887]	0.2686
releff@10	hybrid	within_env	12	0.0362	[0.0014, 0.0854]	1.0000
releff@10	hybrid	family	12	-0.0070	[-0.0293, 0.0143]	1.0000
releff@10	hybrid	cross_env	36	0.0084	[-0.0074, 0.0246]	1.0000
releff@10	ccc	random	12	0.0326	[0.0139, 0.0969]	0.0059
releff@10	ccc	within_env	12	0.0212	[-0.0114, 0.0676]	1.0000
releff@10	ccc	family	12	0.0067	[-0.0090, 0.0247]	1.0000
releff@10	ccc	cross_env	36	-0.0014	[-0.0188, 0.0170]	1.0000

Table S10: Experiment F, per-architecture breakdown. Delta-versus-MSE (affine-calibrated) of Pearson r and NDCG@10 by split scheme and architecture, computed directly from the difficult-splits grid. Bootstrap mean and BCa 95% CI.

Metric	Loss	Scheme	Architecture	n	Mean Δ	95% CI
pearson	pearson	random	mlp	4	0.0041	[0.0004, 0.0080]
pearson	pearson	random	cnn	4	0.0180	[-0.0171, 0.0738]
pearson	pearson	random	transformer	4	0.0254	[0.0129, 0.0344]
pearson	pearson	within_env	mlp	4	-0.0071	[-0.0200, 0.0019]
pearson	pearson	within_env	cnn	4	-0.0162	[-0.0378, 0.0020]
pearson	pearson	within_env	transformer	4	0.0627	[0.0293, 0.0998]
pearson	pearson	family	mlp	4	-0.0046	[-0.0109, 0.0015]
pearson	pearson	family	cnn	4	-0.0168	[-0.0363, 0.0027]
pearson	pearson	family	transformer	4	0.0028	[-0.0151, 0.0208]
pearson	pearson	cross_env	mlp	12	0.0068	[-0.0012, 0.0174]
pearson	pearson	cross_env	cnn	12	-0.0163	[-0.0318, 0.0014]
pearson	pearson	cross_env	transformer	12	0.0038	[-0.0129, 0.0223]
pearson	hybrid	random	mlp	4	0.0040	[0.0009, 0.0072]
pearson	hybrid	random	cnn	4	0.0230	[-0.0112, 0.0735]
pearson	hybrid	random	transformer	4	0.0298	[0.0176, 0.0378]
pearson	hybrid	within_env	mlp	4	-0.0009	[-0.0123, 0.0119]
pearson	hybrid	within_env	cnn	4	-0.0072	[-0.0301, 0.0157]
pearson	hybrid	within_env	transformer	4	0.0645	[0.0328, 0.0971]
pearson	hybrid	family	mlp	4	0.0017	[-0.0046, 0.0084]
pearson	hybrid	family	cnn	4	-0.0116	[-0.0325, 0.0093]
pearson	hybrid	family	transformer	4	0.0095	[-0.0080, 0.0255]
pearson	hybrid	cross_env	mlp	12	0.0022	[-0.0048, 0.0129]
pearson	hybrid	cross_env	cnn	12	-0.0205	[-0.0342, -0.0011]
pearson	hybrid	cross_env	transformer	12	0.0042	[-0.0109, 0.0252]

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Table S10: Experiment F, per-architecture breakdown. Delta-versus-MSE (affine-calibrated) of Pearson r and NDCG@10 by split scheme and architecture, computed directly from the difficult-splits grid. Bootstrap mean and BCa 95% CI.

Metric	Loss	Scheme	Architecture	n	Mean Δ	95% CI
pearson	ccc	random	mlp	4	0.0038	[0.0008, 0.0094]
pearson	ccc	random	cnn	4	0.0374	[0.0056, 0.0905]
pearson	ccc	random	transformer	4	0.0325	[0.0186, 0.0411]
pearson	ccc	within_env	mlp	4	-0.0077	[-0.0231, 0.0052]
pearson	ccc	within_env	cnn	4	0.0278	[0.0141, 0.0447]
pearson	ccc	within_env	transformer	4	0.0617	[0.0259, 0.0975]
pearson	ccc	family	mlp	4	-0.0026	[-0.0095, 0.0081]
pearson	ccc	family	cnn	4	0.0044	[-0.0065, 0.0209]
pearson	ccc	family	transformer	4	0.0086	[-0.0064, 0.0267]
pearson	ccc	cross_env	mlp	12	0.0010	[-0.0088, 0.0110]
pearson	ccc	cross_env	cnn	12	-0.0207	[-0.0389, 0.0035]
pearson	ccc	cross_env	transformer	12	0.0042	[-0.0147, 0.0221]
ndcg@10	pearson	random	mlp	4	0.0012	[-0.0011, 0.0026]
ndcg@10	pearson	random	cnn	4	0.0192	[-0.0050, 0.0591]
ndcg@10	pearson	random	transformer	4	0.0063	[0.0020, 0.0119]
ndcg@10	pearson	within_env	mlp	4	-0.0055	[-0.0141, 0.0032]
ndcg@10	pearson	within_env	cnn	4	-0.0080	[-0.0225, 0.0078]
ndcg@10	pearson	within_env	transformer	4	0.0354	[0.0252, 0.0457]
ndcg@10	pearson	family	mlp	4	-0.0012	[-0.0079, 0.0029]
ndcg@10	pearson	family	cnn	4	-0.0014	[-0.0206, 0.0177]
ndcg@10	pearson	family	transformer	4	0.0106	[-0.0005, 0.0183]
ndcg@10	pearson	cross_env	mlp	12	-0.0031	[-0.0084, 0.0028]
ndcg@10	pearson	cross_env	cnn	12	-0.0054	[-0.0170, 0.0105]
ndcg@10	pearson	cross_env	transformer	12	0.0031	[-0.0135, 0.0179]
ndcg@10	hybrid	random	mlp	4	0.0022	[0.0011, 0.0035]
ndcg@10	hybrid	random	cnn	4	0.0239	[-0.0054, 0.0636]
ndcg@10	hybrid	random	transformer	4	0.0080	[0.0045, 0.0130]
ndcg@10	hybrid	within_env	mlp	4	0.0083	[-0.0029, 0.0195]
ndcg@10	hybrid	within_env	cnn	4	-0.0050	[-0.0136, 0.0060]
ndcg@10	hybrid	within_env	transformer	4	0.0389	[0.0176, 0.0600]
ndcg@10	hybrid	family	mlp	4	-0.0001	[-0.0101, 0.0066]
ndcg@10	hybrid	family	cnn	4	-0.0022	[-0.0235, 0.0191]
ndcg@10	hybrid	family	transformer	4	0.0056	[-0.0094, 0.0205]
ndcg@10	hybrid	cross_env	mlp	12	0.0032	[-0.0037, 0.0108]
ndcg@10	hybrid	cross_env	cnn	12	-0.0065	[-0.0175, 0.0079]
ndcg@10	hybrid	cross_env	transformer	12	0.0079	[-0.0027, 0.0196]
ndcg@10	ccc	random	mlp	4	0.0021	[0.0011, 0.0030]
ndcg@10	ccc	random	cnn	4	0.0268	[0.0020, 0.0653]
ndcg@10	ccc	random	transformer	4	0.0075	[0.0046, 0.0123]
ndcg@10	ccc	within_env	mlp	4	-0.0082	[-0.0144, 0.0001]
ndcg@10	ccc	within_env	cnn	4	0.0039	[-0.0068, 0.0158]

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Table S10: Experiment F, per-architecture breakdown. Delta-versus-MSE (affine-calibrated) of Pearson r and NDCG@10 by split scheme and architecture, computed directly from the difficult-splits grid. Bootstrap mean and BCa 95% CI.

Metric	Loss	Scheme	Architecture	n	Mean Δ	95% CI
ndcg@10	ccc	within_env	transformer	4	0.0428	[0.0186, 0.0670]
ndcg@10	ccc	family	mlp	4	-0.0020	[-0.0022, -0.0017]
ndcg@10	ccc	family	cnn	4	0.0031	[-0.0128, 0.0191]
ndcg@10	ccc	family	transformer	4	0.0135	[0.0004, 0.0267]
ndcg@10	ccc	cross_env	mlp	12	0.0027	[-0.0030, 0.0177]
ndcg@10	ccc	cross_env	cnn	12	-0.0095	[-0.0234, -0.0005]
ndcg@10	ccc	cross_env	transformer	12	0.0090	[-0.0098, 0.0215]

10 Per-dataset results

Table S11 lists, for every (dataset, trait), the sample size, marker count and Pearson predictive ability of GBLUP, ridge, the MSE-trained MLP and the best correlation-loss neural network.

Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:barley	BaMMV	barley	1438	20000	0.649	0.639	0.660	0.658 (mlp/pearson)
easygese:barley	BaYMV	barley	1438	20000	0.511	0.471	0.505	0.509 (mlp/hybrid)
easygese:bean	DF	bean	444	16707	0.743	0.752	0.710	0.753 (cnn/ccc)
easygese:bean	DPM	bean	444	16707	0.666	0.669	0.638	0.641 (mlp/hybrid)
easygese:bean	PHI	bean	444	16707	0.421	0.425	0.389	0.398 (cnn/ccc)
easygese:bean	X100SdW	bean	444	16707	0.648	0.651	0.609	0.626 (mlp/hybrid)
easygese:bean	Yd	bean	444	16707	0.366	0.351	0.319	0.353 (mlp/ccc)
easygese:lentil	DTF	lentil	324	20000	0.798	0.796	0.800	0.804 (mlp/hybrid)
easygese:lentil	DTM	lentil	324	20000	0.828	0.831	0.828	0.830 (mlp/hybrid)
easygese:lentil	DTS	lentil	324	20000	0.844	0.845	0.836	0.847 (mlp/ccc)
easygese:lentil	REP	lentil	324	20000	0.682	0.682	0.673	0.693 (mlp/ccc)
easygese:lentil	VEG	lentil	324	20000	0.819	0.819	0.827	0.828 (mlp/ccc)
easygese:maize	Ear_ Height_ cm	maize	4420	20000	0.819	0.806	0.814	0.814 (mlp/hybrid)
easygese:maize	Grain_ Moisture	maize	4421	20000	0.876	0.874	0.879	0.879 (mlp/hybrid)
easygese:maize	Plant_ Height_ cm	maize	4420	20000	0.841	0.831	0.839	0.840 (mlp/pearson)

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Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:maize	Pollen_DAP_days	maize	4416	20000	0.926	0.922	0.925	0.926 (mlp/ccc)
easygese:maize	Silk_DAP_days	maize	4413	20000	0.926	0.922	0.922	0.923 (mlp/pearson)
easygese:maize	Yield_Mg_ha	maize	4421	20000	0.585	0.584	0.638	0.637 (mlp/hybrid)
easygese:oyster	daydead	oyster	214	20000	0.299	0.275	0.223	0.302 (cnn/ccc)
easygese:oyster	mm	oyster	372	20000	0.457	0.459	0.406	0.434 (cnn/ccc)
easygese:pig	MS	pig	1709	20000	0.321	0.285	0.310	0.315 (mlp/ccc)
easygese:pig	PFAI	pig	1709	20000	0.291	0.257	0.275	0.287 (mlp/ccc)
easygese:pine	BA	pine	861	4421	0.811	0.812	0.802	0.806 (mlp/ccc)
easygese:pine	BD	pine	861	4421	0.811	0.816	0.816	0.820 (mlp/pearson)
easygese:pine	BLC	pine	861	4421	0.791	0.789	0.784	0.790 (mlp/pearson)
easygese:pine	CWAC	pine	861	4421	0.711	0.714	0.700	0.705 (mlp/ccc)
easygese:pine	CWAL	pine	861	4421	0.762	0.769	0.750	0.760 (mlp/hybrid)
easygese:pine	DBH	pine	861	4421	0.714	0.718	0.714	0.719 (mlp/hybrid)
easygese:pine	HT	pine	861	4421	0.611	0.620	0.609	0.614 (mlp/pearson)
easygese:pine	HTLC	pine	861	4421	0.786	0.784	0.775	0.778 (mlp/pearson)
easygese:pine	c5c6	pine	910	4421	0.891	0.906	0.901	0.907 (mlp/pearson)
easygese:pine	density	pine	910	4421	0.934	0.944	0.939	0.942 (mlp/pearson)
easygese:pine	gall	pine	807	4421	0.889	0.890	0.887	0.889 (mlp/ccc)
easygese:pine	lateWood.4	pine	910	4421	0.894	0.905	0.901	0.903 (mlp/pearson)
easygese:pine	lignin	pine	910	4421	0.893	0.905	0.904	0.907 (mlp/pearson)
easygese:pine	rootnum	pine	925	4421	0.866	0.871	0.868	0.874 (mlp/hybrid)
easygese:pine	rootnumbin	pine	925	4421	0.846	0.856	0.859	0.863 (mlp/pearson)
easygese:pine	rustbin	pine	807	4421	0.848	0.854	0.845	0.849 (mlp/ccc)
easygese:pine	stiffnessTree	pine	910	4421	0.864	0.868	0.859	0.866 (mlp/ccc)
easygese:rice	Alkali_spreading_value	rice	345	20000	0.285	0.276	0.244	0.328 (cnn/ccc)
easygese:rice	Amylose_content	rice	344	20000	0.658	0.659	0.640	0.675 (cnn/pearson)

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Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:rice	Awn_ presence	rice	338	20000	0.169	0.197	0.129	0.180 (cnn/pearson)
easygese:rice	Blast_ resistance	rice	342	20000	0.514	0.510	0.457	0.506 (cnn/ccc)
easygese:rice	Brown_ rice_ length_ width_ ratio	rice	346	20000	0.622	0.620	0.546	0.617 (cnn/ccc)
easygese:rice	Brown_ rice_ seed_ length	rice	346	20000	0.546	0.542	0.445	0.556 (cnn/ccc)
easygese:rice	Brown_ rice_ seed_ width	rice	346	20000	0.664	0.667	0.628	0.677 (cnn/hybrid)
easygese:rice	Brown_ rice_ surface_ area	rice	346	20000	0.507	0.518	0.451	0.573 (cnn/pearson)
easygese:rice	Brown_ rice_ volume	rice	346	20000	0.600	0.605	0.579	0.640 (cnn/hybrid)
easygese:rice	Culm_ habit	rice	351	20000	0.607	0.607	0.572	0.598 (cnn/hybrid)
easygese:rice	FT_ ratio_ of_ Arkansas_ Aberdeen	rice	319	20000	0.351	0.372	0.258	0.309 (cnn/ccc)
easygese:rice	FT_ ratio_ of_ Faridpur_ Aberdeen	rice	280	20000	0.153	0.261	0.203	0.238 (cnn/ccc)
easygese:rice	Flag_ leaf_ length	rice	346	20000	0.323	0.333	0.233	0.310 (cnn/ccc)
easygese:rice	Flag_ leaf_ width	rice	346	20000	0.640	0.639	0.608	0.667 (cnn/ccc)
easygese:rice	Florets_ per_ panicle	rice	346	20000	0.417	0.426	0.329	0.422 (cnn/pearson)

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Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:rice	Flowering_time_at_Aberdeen	rice	325	20000	0.011	0.144	0.159	0.244 (cnn/ccc)
easygese:rice	Flowering_time_at_Arkansas	rice	342	20000	0.449	0.449	0.393	0.485 (cnn/ccc)
easygese:rice	Flowering_time_at_Faridpur	rice	281	20000	0.339	0.341	0.294	0.330 (mlp/hybrid)
easygese:rice	Leaf_pubescence	rice	265	20000	0.446	0.451	0.357	0.426 (cnn/hybrid)
easygese:rice	Panicle_fertility	rice	345	20000	0.345	0.345	0.290	0.342 (cnn/ccc)
easygese:rice	Panicle_length	rice	344	20000	0.522	0.525	0.434	0.505 (cnn/pearson)
easygese:rice	Panicle_number_per_plant	rice	341	20000	0.734	0.736	0.712	0.741 (cnn/ccc)
easygese:rice	Pericarp_color	rice	346	20000	0.461	0.373	0.369	0.487 (cnn/pearson)
easygese:rice	Plant_height	rice	352	20000	0.605	0.600	0.531	0.610 (cnn/ccc)
easygese:rice	Primary_panicle_branch_number	rice	344	20000	0.474	0.474	0.396	0.496 (cnn/hybrid)
easygese:rice	Protein_content	rice	338	20000	0.370	0.378	0.261	0.353 (cnn/ccc)
easygese:rice	Seed_color	rice	346	20000	0.094	0.057	0.269	0.331 (mlp/hybrid)
easygese:rice	Seed_length	rice	346	20000	0.478	0.478	0.377	0.450 (cnn/pearson)
easygese:rice	Seed_length_width_ratio	rice	346	20000	0.603	0.603	0.549	0.588 (cnn/ccc)
easygese:rice	Seed_number_per_panicle	rice	345	20000	0.294	0.329	0.312	0.336 (cnn/pearson)

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Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:rice	Seed_surface_area	rice	346	20000	0.494	0.503	0.408	0.529 (cnn/hybrid)
easygese:rice	Seed_volume	rice	346	20000	0.585	0.591	0.536	0.615 (cnn/hybrid)
easygese:rice	Seed_width	rice	346	20000	0.654	0.656	0.614	0.667 (transformer/ccc)
easygese:rice	Straighthead_suseptability	rice	304	20000	0.592	0.595	0.522	0.560 (cnn/ccc)
easygese:rice	Year06Flowering_time_at_Arkansas	rice	308	20000	0.487	0.489	0.387	0.491 (cnn/ccc)
easygese:rice	Year07Flowering_time_at_Arkansas	rice	337	20000	0.435	0.431	0.394	0.461 (cnn/ccc)
easygese:soybean	C13	soybean	346	20000	0.379	0.370	0.357	0.365 (mlp/ccc)
easygese:soybean	Wilt	soybean	346	20000	0.418	0.421	0.389	0.414 (mlp/hybrid)
soynam:height	height	soybeanNAM	5487	4401	0.680	0.679	0.677	0.683 (mlp/hybrid)
soynam:oil	oil	soybeanNAM	5487	4401	0.715	0.715	0.703	0.713 (mlp/hybrid)
soynam:protein	protein	soybeanNAM	5487	4401	0.732	0.733	0.726	0.727 (mlp/ccc)
soynam:yield	yield	soybeanNAM	5487	4401	0.708	0.700	0.695	0.696 (mlp/pearson)
wheat	env1	wheat	599	1279	0.511	0.514	0.546	0.548 (mlp/hybrid)
wheat	env2	wheat	599	1279	0.494	0.496	0.453	0.458 (mlp/hybrid)
wheat	env3	wheat	599	1279	0.388	0.386	0.379	0.368 (mlp/hybrid)
wheat	env4	wheat	599	1279	0.431	0.428	0.453	0.465 (cnn/ccc)
easygese:wheatG	DTR	wheatG	371	12546	0.441	0.443	0.397	0.415 (mlp/ccc)
easygese:wheatG	EW	wheatG	371	12546	0.369	0.374	0.328	0.337 (mlp/ccc)
easygese:wheatG	FHB	wheatG	371	12546	0.700	0.706	0.700	0.707 (mlp/ccc)
easygese:wheatG	GH	wheatG	371	12546	0.573	0.571	0.558	0.574 (mlp/hybrid)
easygese:wheatG	GPE	wheatG	371	12546	0.341	0.339	0.301	0.332 (mlp/hybrid)
easygese:wheatG	GY	wheatG	371	12546	0.629	0.627	0.601	0.608 (mlp/hybrid)
easygese:wheatG	HAG	wheatG	371	12546	0.466	0.466	0.442	0.454 (mlp/pearson)

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Table S11: Per (dataset, trait) summary. Test-set Pearson r (mean over cross-validation folds) for GBLUP (raw), ridge (affine), the MSE-trained MLP (affine) and the best correlation-loss neural network (affine; winning model/loss in parentheses). n is the maximum train+test sample count and p the number of markers used. Sorghum is excluded (proprietary).

Dataset	Trait	Species	n	p	GBLUP	Ridge	MLP/MSE	Best corr. NN
easygese:wheatG	HD	wheatG	371	12546	0.672	0.690	0.700	0.698 (mlp/pearson)
easygese:wheatG	PC	wheatG	371	12546	0.658	0.667	0.650	0.665 (mlp/ccc)
easygese:wheatG	PH	wheatG	371	12546	0.795	0.803	0.805	0.805 (mlp/ccc)
easygese:wheatG	SDS	wheatG	371	12546	0.757	0.767	0.747	0.741 (mlp/hybrid)
easygese:wheatG	SEP	wheatG	371	12546	0.490	0.492	0.463	0.472 (mlp/pearson)
easygese:wheatG	STC	wheatG	371	12546	0.574	0.580	0.561	0.542 (mlp/pearson)
easygese:wheatG	SW	wheatG	371	12546	0.743	0.745	0.744	0.749 (mlp/ccc)
easygese:wheatG	TKW	wheatG	371	12546	0.430	0.442	0.415	0.432 (mlp/ccc)
easygese:wheatG	ZEL	wheatG	371	12546	0.775	0.778	0.760	0.760 (mlp/ccc)

11 Supplementary figures

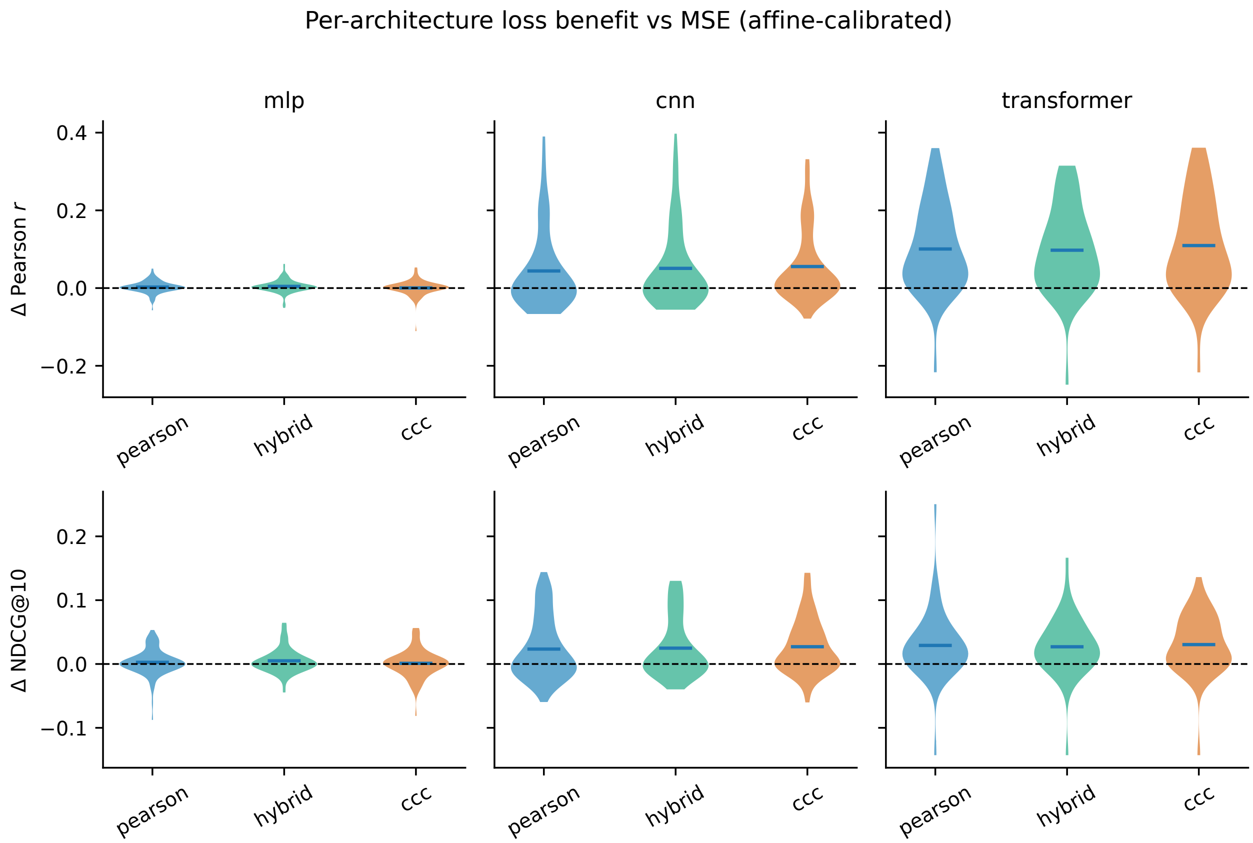

Figure S1: Per-architecture distributions of Δ Pearson r (top) and Δ NDCG@10 (bottom) for the Pearson, hybrid and CCC losses relative to MSE, for the affine-calibrated MLP, CNN and Transformer. Violins show the distribution over (dataset, trait) cells; the dashed line marks no change. The benefit is substantial for the CNN and Transformer and small for the well-tuned MLP.

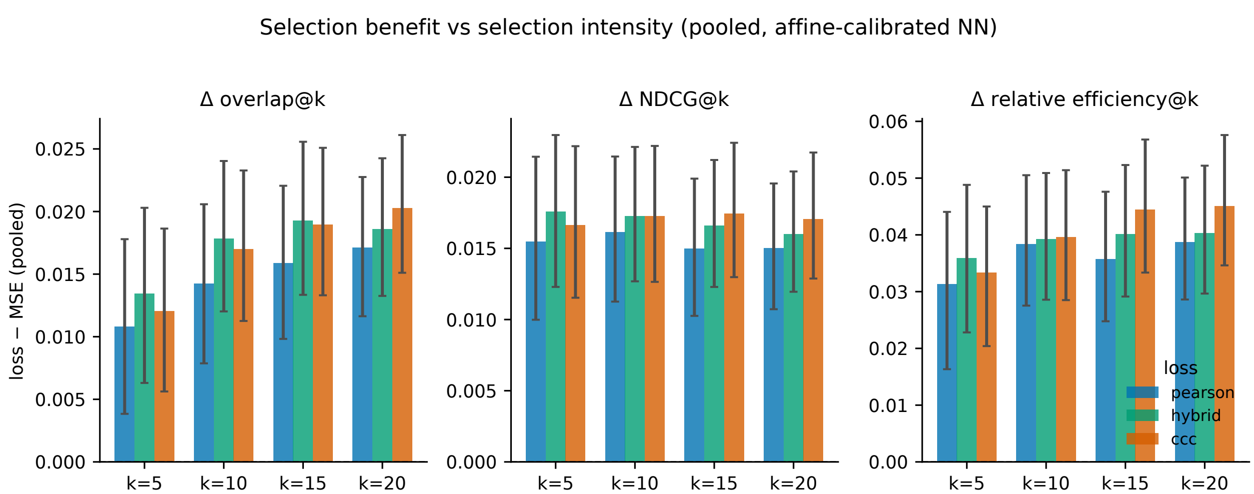

Figure S2: Pooled Δ of three selection metrics — top-k overlap, NDCG@k and relative efficiency@k — at selection sizes $k \in \{5, 10, 15, 20\}$, for the affine-calibrated neural network. Bars are bootstrap means with BCa 95% confidence intervals; the dashed line marks no change. The selection advantage of correlation-consistent training is present across selection intensities.

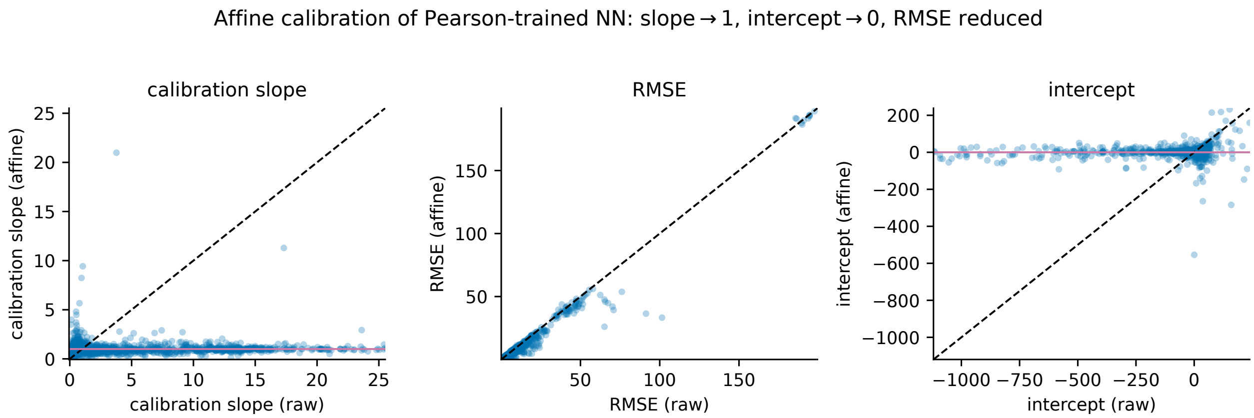

Figure S3: Affine calibration of the Pearson-trained neural network (extends main text Fig. 5). Each point is a cross-validation fold; raw values on the x-axis, affine-calibrated values on the y-axis, for the calibration slope, RMSE and intercept. Calibration drives the slope towards 1 and the intercept towards 0 and reduces RMSE while leaving the rank correlation unchanged.

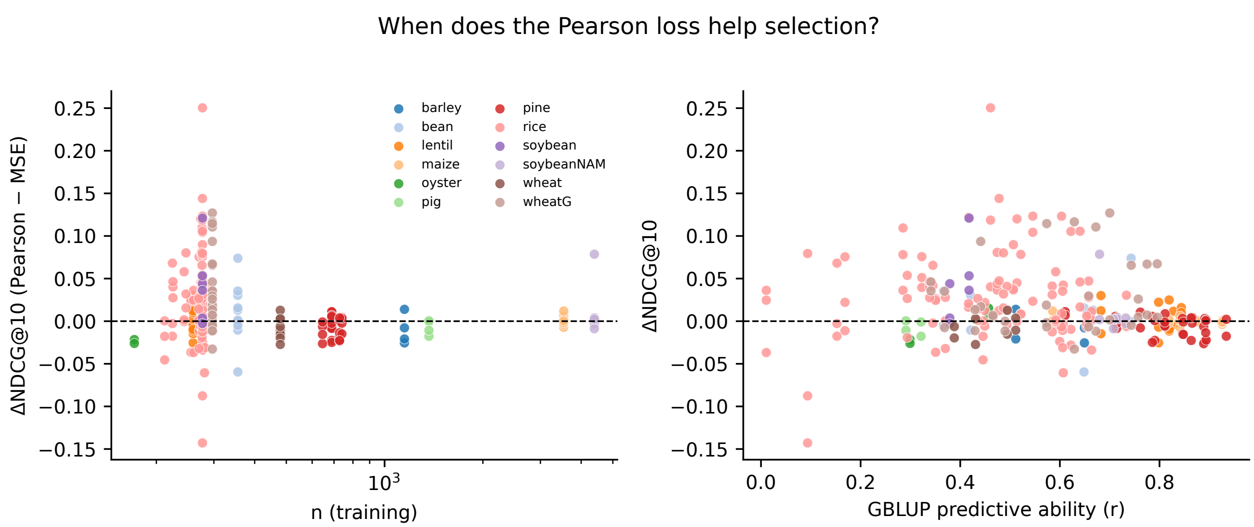

Figure S4: Meta-analysis of when correlation-consistent training helps selection. Δ NDCG@10 of the Pearson loss versus MSE plotted against training-set size (left) and against GBLUP predictive ability (right), colored by species, for the affine-calibrated neural network.

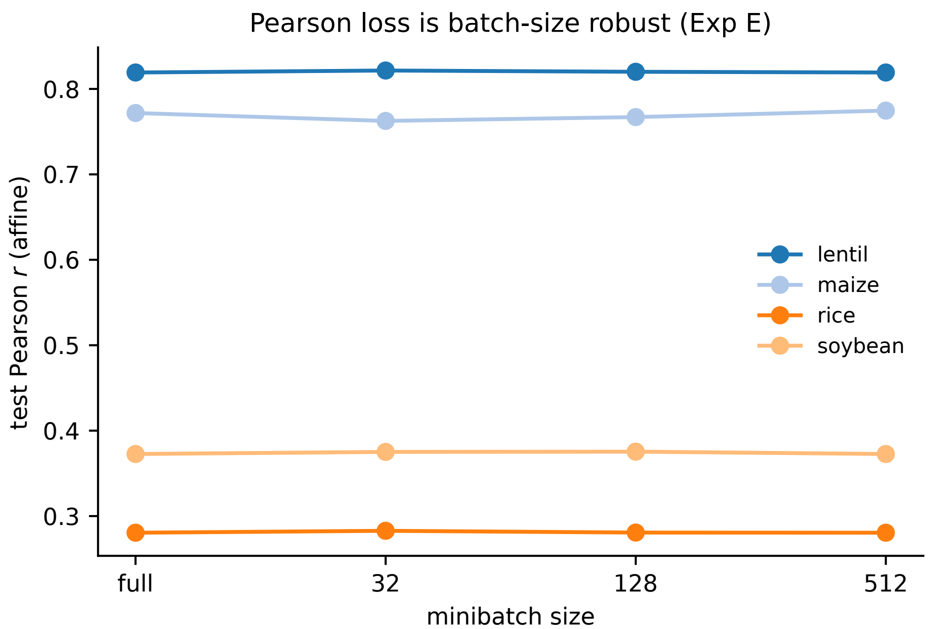

Figure S5: Experiment E: test Pearson r (affine-calibrated MLP, Pearson loss) versus minibatch size for each of four EasyGeSe datasets ("full" = full-batch gradient). Accuracy is insensitive to the minibatch size.